

Federated Hermes Prompt Engineering Guide

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This document is designed to introduce you to the concept of prompt engineering, or writing an effective prompt, and explain why it's an important skill when working with Generative AI models like Microsoft Copilot. By following this guide, you will learn how to write effective prompts, understand the capabilities and limitations of Copilot, and apply it to various business scenarios to enhance productivity and innovation.

Key Principles for Effective AI Use

Before we get started, let's consider a few guiding principles that will help you get the most out of the AI tools that are becoming more accessible in our day-to-day work environment.

Invite AI to a Seat at Your Table

Think of AI as a valuable team member. Integrate it into your workflow and use it as a resource for brainstorming, problem-solving, and generating new ideas. However, remember that AI is a tool to assist you, not replace human judgment.

Be the Human in the Loop

Always maintain human oversight. For instance, if Copilot drafts an investment report, you must review and validate its output to ensure accuracy and relevance paying particular attention to numeric values, measures, or size-based comparisons. This ensures that the final product meets the high standards required in our industry.

Treat the AI as a Person, but Tell It What Kind of Person It Is

When interacting with AI, provide clear instructions and context. For example, specify the role or perspective you want the AI to adopt, such as an investment analyst, marketing strategist, or citizen developer. This helps the AI generate more relevant and accurate responses.

AI Is Not Traditional Software

Unlike traditional software, AI is not entirely predictable. Its responses can vary based on the input and context. Furthermore, while responses for a given prompt may be generally consistent, they too may vary each time you submit the prompt. This is the nature of Generative AI models. Embrace this flexibility and use it to your advantage by experimenting with different prompts and refining your queries.

Approach with Cautious Optimism

While AI is continually improving, it's essential to manage expectations. Consider the AI's current capabilities as a starting point, knowing AI will continue to get better. This mindset encourages you to critically evaluate AI outputs and continuously seek ways to optimize their use.

Know When to Use the Right Tool

For specific, fact-based, or real-time queries, use a search engine or a trusted database. Reserve tools like Microsoft Copilot for tasks that require creativity, summarization, or structured assistance, rather than exact data retrieval.

When to Use Generative AI Tools like Microsoft Copilot

Generative AI tools like Microsoft Copilot are powerful resources that can assist business users in a wide variety of tasks, enabling greater productivity, efficiency, and innovation. However, like any tool, they are most effective when used for the right purposes and with an understanding of their limitations. Below, we outline some of the best use cases for these tools, as well as scenarios where caution or alternative approaches are more appropriate.

When to Use Microsoft Copilot

Generative AI tools excel in areas where creativity, efficiency, and textual data processing intersect. You can leverage these tools to streamline workflows, spark ideas, and reduce manual effort. Here are some example practical use cases:

- **Content Creation and Drafting:**
 - Drafting emails, reports, proposals, or presentations.
 - Generating creative content, such as marketing copy, headlines, or social media posts.
 - Summarizing long documents or meeting notes into concise, actionable points.
- **Idea Generation and Brainstorming:**
 - Generating ideas for campaigns, product names, or project plans.
 - Exploring alternative approaches or solutions to a business challenge.
 - Providing creative inspiration to jump-start new initiatives.
- **Data Analysis Support:**
 - Interpreting data trends or summarizing reports in plain language.
 - Creating visualizations (e.g., Excel charts) or turning raw data into insights.
 - Assisting in drafting SQL queries or formulas for data management tasks.
- **Task Automation and Productivity:**
 - Automating repetitive tasks like formatting text, proofreading, or rephrasing content.
 - Providing templates or frameworks for documents, workflows, or processes.
 - Generating checklists, timelines, or project plans based on simple inputs.
- **Learning and Knowledge Exploration:**
 - Answering questions to quickly research unfamiliar topics.
 - Explaining complex concepts in a simplified manner.
 - Offering step-by-step guides for new tools, platforms, or methodologies.

When to Consider Other Options

While Generative AI tools are remarkably versatile, they are not infallible and have limitations that you should be aware of. Below are some example scenarios where Copilot may not be the best option or where users should exercise caution:

- **Critical Decision-Making:**
 - Generative AI tools are not substitutes for human judgment, particularly in high-stakes or strategic decisions.
 - Use outputs as suggestions or starting points, but validate them with domain expertise and critical analysis.
- **Generating Factual or Legal Content:**
 - The models are prone to "hallucinations," where they may provide incorrect or made-up information that sounds plausible.
 - Avoid relying on them for precise facts, legal advice, or compliance-related content without verifying the information.
- **Specialized or Technical Expertise:**
 - While the tools can simulate expertise in many areas, they lack true subject matter understanding and may provide incomplete or inaccurate responses for highly technical or specialized tasks.
- **Complex Multistep Processes Requiring Context Retention:**
 - Generative AI tools can struggle with maintaining context over long or complex interactions, which may lead to inconsistencies or errors.
 - For intricate workflows, break tasks into smaller steps or use specialized tools designed for those processes.
- **Tasks Requiring Emotional Intelligence or Nuance:**
 - Use AI for drafting or suggestion purposes, but ensure human oversight for tone-sensitive communications.
- **Replacing Traditional Search Engines for Factual Queries:**
 - Generative AI tools like Microsoft Copilot are not optimized for retrieving exact, up-to-date, or comprehensive information in the same way a search engine (e.g., Google or Bing) is.
 - Even though Microsoft Copilot may integrate with live search functionality, it often summarizes information from multiple sources rather than providing a definitive, verified answer.
 - Generative AI models can sometimes "hallucinate," generating plausible but incorrect or fabricated information, which makes them unreliable for fact-based research.

Getting Started with Generative AI

As you begin to leverage AI in your work processes, consider these recommended practices:

- **Start Simple:** Begin with straightforward tasks and gradually move to more complex ones as you become more comfortable with the tool.
- **Iterate and Refine:** Continuously refine your prompts based on the AI's responses. Iterative improvement is key to effective prompt engineering.
- **Leverage Advanced Techniques:** As you gain confidence, explore advanced strategies such as Chain of Thought, Few-Shot Learning, and Contextual Priming to enhance the quality of AI-generated outputs.

Generative AI is all about the prompts, so to begin:

What are Prompts?

In the context of Copilot, a prompt is the input or question you provide to the model to elicit a response. It can be a single sentence, a question, or a detailed set of instructions. The quality of the prompt directly influences the relevance and accuracy of the response generated by Copilot. Effective prompts are clear, concise, and provide sufficient context to guide the model towards producing the desired output.

What is Prompt Engineering?

Prompt Engineering is the practice of designing and refining prompts to maximize the quality and usefulness of responses from Generative AI models like Copilot. It involves understanding how to phrase questions, provide context, and structure information to guide the model's responses effectively. Prompt engineering is a critical skill for leveraging Copilot, ensuring that the interactions are productive and aligned with your objectives, as well as guiding the AI to deliver more accurate and contextually relevant outputs.

Why is Prompt Engineering Important?

Generative AI relies heavily on the quality and clarity of the input it receives. The more precisely and contextually a task is described, the more effectively the AI can respond. Developing the skill of prompt engineering allows users to craft queries that more effectively harness the AI's capabilities. This skill is essential for a wide range of applications, from generating creative text and producing images to even writing code.

Understanding Microsoft Copilot

Copilot is a powerful Generative AI model developed by Microsoft and OpenAI that can perform a wide range of tasks, including but not limited to:

- **Text / Image Generation:** Creating coherent and contextually relevant text or images based on a given prompt. This can be used for drafting emails, writing reports, generating content for social media, creating images for presentations and more.
- **Question Answering:** Responding to factual questions or providing explanations on various topics.
- **Summarization:** Condensing long pieces of text into concise summaries, which is useful for quickly understanding large documents or articles.
- **Translation:** Translating text between different languages.
- **Conversation Simulation:** Engaging in human-like conversations for customer support, training, or other interactive applications.
- **Creative Writing:** Assisting in writing creative content.
- **Coding Assistance:** Helping with code generation, debugging, and offering programming advice.

While Copilot is highly capable, it has certain limitations that users should be aware of:

- **Accuracy:** Copilot may sometimes produce incorrect or nonsensical answers. It is important to verify the information provided, especially for critical tasks.
- **Context Retention:** The model can lose context over long conversations or complex interactions, which may require users to reestablish context periodically.
- **Bias:** Copilot can inadvertently reflect biases present in the training data. Users should be cautious and critically evaluate the responses for fairness and inclusivity.
- **Sensitivity to Input:** Small changes in the input prompt can lead to significantly different responses. Crafting precise and clear prompts is essential.
- **Creativity vs. Factuality:** While Copilot can generate creative content, it may not always distinguish between factual information and plausible-sounding but incorrect details.

Best Practices for Using Microsoft Copilot

- **Be Clear and Specific:** Provide clear, specific prompts to guide the model effectively.
- **Iterate and Refine:** Start with a basic prompt and refine it based on the responses you receive.
- **Maintain Context:** When engaging in extended interactions, periodically remind the model of the context to maintain coherence.
- **Review and Edit:** Always review and edit the AI-generated content to ensure accuracy and relevance.
- **Use Feedback:** Provide feedback on the responses to help improve future interactions.

Strategies for Crafting Effective Prompts

Crafting effective prompts involves understanding the basic structure and elements that make up a good prompt. Here are the key components:

- **Clarity:** The prompt should be clear and easy to understand. Avoid ambiguous language and be precise about what you are asking.
- **Context:** Provide enough background information to help the model understand the situation or topic. This could include previous interactions, specific details, or relevant data.
- **Specificity:** Be specific about what you want. The more detailed and focused the prompt, the better the model can tailor its response.

Avoiding common mistakes can improve the quality of your prompts. Here are some pitfalls to watch out for:

- **Vagueness:** Prompts that are too broad or vague can lead to irrelevant or incomplete responses. Be as specific as possible.
- **Lack of Context:** Not providing enough background information can result in the model making incorrect assumptions. Always include relevant details.
- **Overloading:** Including too many questions or requests in a single prompt can confuse the model. Break down complex queries into simpler, individual prompts.

Here are some additional tips to help you craft more effective prompts:

Be Specific and Clear

When crafting prompts, specificity and clarity are essential. Simple and direct language often yields the best results. Vague prompts can lead to ambiguous or irrelevant responses. Generative AI only responds to the instructions you provide, so clear, detailed prompts will yield better results.

Example:

- **Vague:** "Tell me about marketing."
- **Specific:** "Explain the key components of a digital marketing strategy, including SEO, content marketing, and social media."

Provide Context (Contextual Priming)

Without sufficient context, the AI may make incorrect assumptions or provide incomplete responses. Giving context can significantly improve the quality of responses. Include relevant background information, instructions or guidelines to guide the model.

Example:

- **Without Context:** "What should I do next?"
- **With Context:** "I just completed the attached investment management research report, which highlights the performance of various asset classes over the last quarter. What should be the next steps in adjusting our investment portfolio?"

Set the Role or Persona (Role-Playing)

Assigning a role or persona to the model can help tailor responses to your needs. This is particularly useful for simulating specific scenarios or professional interactions.

Example:

- **Role Assignment:** "You are a sales rep for a large asset management company. How would you respond to a client who is concerned about the recent volatility in the stock market and is considering moving their investments to safer assets?"

Use Examples in Prompts (Few-Shot Learning)

Including a few examples of the task you want the model to perform within the prompt helps the model understand the pattern and context, thereby generating more relevant responses.

Example:

- **Without Example:** "Write a social media post about our new Large Cap ETF product."
- **With Example:** " Write a social media post about our new quantitative model driven Large Cap ETF. For example, here is a recent post for one of our other ETF products - 'We're thrilled to announce the launch of our new Environmental Stewardship ETF! Invest in a sustainable future with us. Learn more on our website.'"

Ask for Structured Outputs

If you need the response in a specific format, explicitly ask for it. This can be in the form of lists, bullet points, or specific sections.

Example:

- **General Prompt:** "Can you summarize Disney's Annual Report?"
- **Structured Prompt:** "In bullet point format, list the 3 most important risk factors in Disney's 2024 Annual Report. For each bullet point, provide impacts an investment analyst should be considering."

Give the Model Time to "Think"

Encouraging the model to take a moment to "think" can lead to more thoughtful and comprehensive responses. This can be achieved by explicitly asking the model to consider the question carefully or by structuring the prompt in a way that implies a need for a more detailed response. Additionally, asking for a "chain of thought" before an answer can help the model reason its way toward correct answers more reliably.

Example:

- **Immediate Prompt:** "List three ways to improve our customer service."
- **Thoughtful Prompt:** "Take a moment to consider and then list three detailed strategies for improving our customer service, explaining the benefits of each strategy."

Experiment and Iterate (Iterative Refinement)

How you phrase your prompt can change the outcome. Experimenting with different phrasings and iterating based on the responses can help refine prompts to get the best results. After receiving a response, try adjusting your prompt to get closer to the result you're looking for.

Example:

- **Initial Prompt:** "Give me ideas for a marketing campaign for a product."
- **Refined Prompt:** "Provide five creative ideas for a digital marketing campaign to promote our new fixed income investment ETF product in an environment of lowering interest rates."

After receiving the response, you can further refine:

- Great, now let's focus on the first one and refine to include the following client segmentation as the target audience for the campaign.

Split Complex Tasks into Simpler Subtasks (Chain of Thought)

Breaking down complex tasks into simpler subtasks can help the model focus on one aspect at a time, leading to more accurate and manageable responses.

Example:

- **Complex Task:** "Create a comprehensive marketing plan for our new ETF product launch including key objectives, identification of target audience and suggested marketing channels."
- **Simpler Subtasks:** "I would like to create a comprehensive marketing plan for our new ETF product launch. This plan should:
 1. Outline the key objectives for the marketing plan.
 2. Identify the target audience for the new product.
 3. Suggest marketing channels to use for reaching the target audience.

Let's start with "Outline the key objectives for the marketing plan." Once I review and confirm it is what is expected, we will move on to the next section."

Final Thoughts

- **Practice and Experiment:** The more you use AI and experiment with different prompts, the better you'll become at crafting effective prompts.
- **Continuous Learning:** Stay updated with new techniques and advancements in AI to keep improving your skills.
- **Collaboration:** Share your experiences and successful prompt strategies with colleagues to foster a collaborative learning environment.

Appendix: FAQ and Example Prompts

Despite the powerful capabilities of Copilot, users may encounter some common issues. Here are typical problems and solutions:

Incorrect or Irrelevant Responses

- **Issue:** The model provides answers that are incorrect or not relevant to the prompt.
- **Solution:**
 - Ensure that your prompt is clear, specific, and includes sufficient context.
 - Refine your prompt iteratively to guide the model towards the desired response.
 - Use examples within the prompt to clarify what you are looking for.

Loss of Context in Extended Conversations

- **Issue:** The model loses track of the context in long conversations.
- **Solution:**
 - Periodically restate key points or context to remind the model of the ongoing discussion.
 - Break down long conversations into smaller, more manageable interactions.

Biased or Inappropriate Responses

- **Issue:** The model generates biased, inappropriate, or offensive content.
- **Solution:**
 - Use clear and specific language in prompts to minimize ambiguity.
 - If you encounter inappropriate content, use feedback mechanisms to report it.

Incomplete Responses

- **Issue:** The model provides incomplete answers or cuts off mid-sentence.
- **Solution:**
 - Prompt the model to continue from where it left off. For example, "Please continue the previous response."

Difficulty Handling Complex Queries

- **Issue:** The model struggles to handle complex or multi-part queries.
- **Solution:**
 - Split complex tasks into simpler subtasks and address each one individually.
 - Use chaining prompts to build on previous responses and tackle complex queries step-by-step.

Example 1 – Prompt Engineering for Learning About IM Concepts

Generative AI tools can be extremely useful when learning about new concepts, especially when high quality prompts are used to elicit high quality results. Below is another example that displays how important prompt engineering is when learning about new topics.

In this example, a new Investment Analyst is tasked with learning more about Money Market instruments. The Investment Analyst initially uses the response below:

“Tell me the definition of a money market fund”

A money market fund (MMF) is a type of mutual fund that primarily invests in short-term, high-quality debt securities and cash equivalents. These can include instruments like U.S. Treasury bills, commercial paper, and certificates of deposit. The main goal of a money market fund is to provide investors with high liquidity and a low level of risk, making it a popular choice for those looking to park their cash temporarily while earning some interest.

As you can see, the output of this prompt is vague, and lacks the detail necessary for the Investment Analyst to learn about this concept. Using the strategies discussed in this document, we can modify the prompt to the following:

“As a Senior Investment Analyst, please provide four bullet points that explain what a money market fund is. Include its definition, investment characteristics, liquidity and accessibility, and average returns. Also, provide a real-world example to illustrate these concepts. I am a new Investment Analyst who has no knowledge of money market funds”

Money Market Fund Overview

- **Definition:** A money market fund (MMF) is a type of mutual fund that invests in short-term, high-quality debt instruments, such as U.S. Treasury bills, commercial paper, and certificates of deposit. These funds aim to provide investors with a safe place to park cash while earning a modest return ¹.
- **Investment Characteristics:** MMFs typically invest in low-risk, highly liquid securities with short maturities (usually less than one year). They are designed to maintain a stable net asset value (NAV) of \$1 per share, making them less volatile than other investment options ². The income generated can be taxable or tax-exempt, depending on the fund's investments ³.
- **Liquidity and Accessibility:** Money market funds offer high liquidity, allowing investors to withdraw funds quickly, often with same-day or next-day access. Unlike bank accounts, MMFs are not FDIC-insured but are regulated by the SEC, which mandates strict investment guidelines to ensure safety ⁴. Investors can typically redeem shares without penalties, although some funds may impose liquidity fees during times of market stress ⁵.
- **Average Returns:** The average yield for money market funds is currently around 5.13%, significantly higher than traditional savings accounts, which often yield less than 1% ⁶. Returns can fluctuate based on prevailing interest rates, but MMFs generally provide better returns than standard bank deposits while maintaining low risk ⁷.

Real-World Example

A well-known example of a money market fund is the **Vanguard Federal Money Market Fund (VMFXX)**. This fund primarily invests in U.S. government securities and repurchase agreements, maintaining a low expense ratio of 0.11% and currently offering a 5.28% seven-day SEC yield ⁸. It exemplifies the characteristics of safety, liquidity, and competitive returns that define money market funds.

Feel free to ask if you have more questions or need further details on any specific aspect!

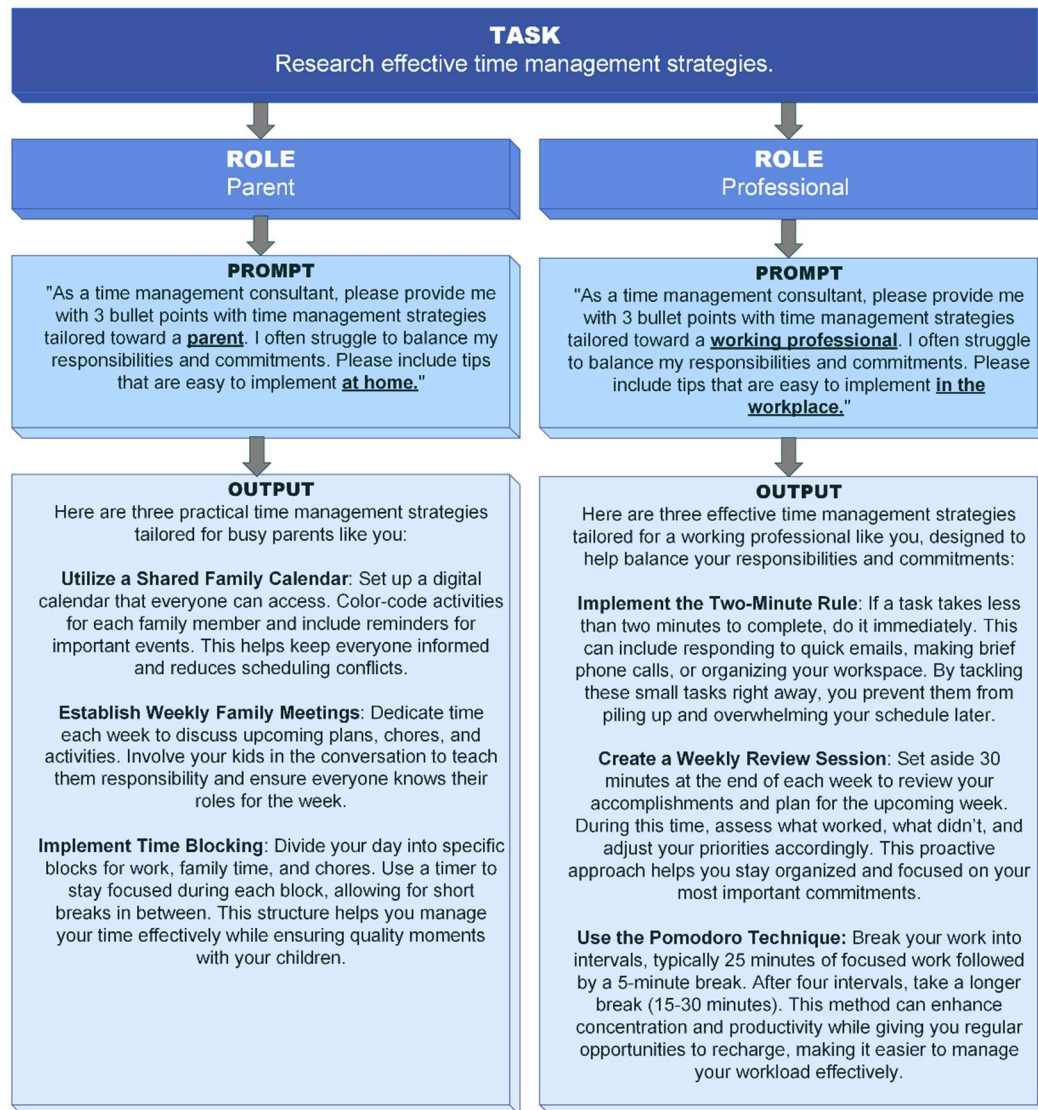
1 7 Best Money Market Fun... 2 10 Best Money Market Mu... 3 Best Money Market Funds... +5

As you can see, this response is much more detailed and provides the information necessary for an entry-level Investment Analyst to learn more about money market funds. The response provided all of the information requested in the prompt, and even provides additional resources at the end if the Investment Analyst would like to research further.

Example 2 – Importance of Context in Prompt Engineering

Next, we will see how providing context when using Generative AI can impact the output of a response. By including relevant details in your prompts, you can receive more accurate and helpful responses tailored to your needs. In this example, we will explore time management strategies for different roles by changing the context of the prompt with only a few words.

As you can see in the example above, the only differences between the two prompts are the underlined words. By providing context, the Generative AI model was able to tailor its advice to the requestor. As you provide more context in your prompt, the output you receive will become increasingly accurate.



Example 3 – Asset Allocation for a Financial Portfolio

Next, we will see how providing additional context can impact the output & increase the value of the information returned:

Initial Prompt: For my financial portfolio, what asset allocation should I have?

For my financial portfolio, what asset allocation should I have?

Determining the right asset allocation for your financial portfolio depends on several factors, including your financial goals, risk tolerance, and investment time horizon. Here are some general guidelines to help you get started:

1. **Identify Your Financial Goals:** Are you saving for retirement, a down payment on a house, or your child's education? Your goals will influence your asset allocation.
2. **Assess Your Risk Tolerance:** How comfortable are you with market fluctuations? If you can tolerate higher risk, you might allocate more to stocks. If you prefer stability, you might lean towards bonds and cash.
3. **Determine Your Time Horizon:** The length of time you plan to invest before needing the money is crucial. Longer time horizons can typically handle more risk.

Here are some common asset allocation models based on different risk tolerances:

- **Conservative Portfolio:** Typically includes a higher percentage of bonds and cash, with a smaller allocation to stocks. This is suitable for those with low risk tolerance or shorter time horizons.
- **Balanced Portfolio:** A mix of stocks and bonds, aiming for moderate growth while managing risk. This is ideal for investors with a medium risk tolerance and a mid- to long-term investment horizon.
- **Aggressive Portfolio:** Primarily composed of stocks, aiming for higher growth but with higher risk. Suitable for those with high risk tolerance and a long-term investment horizon ¹ ².

This initial prompt is very general, so the result is also very general and not as helpful as we wanted it to be. Let's update the prompt:

Updated Prompt: Acting as a financial advisor, provide a breakdown for my financial portfolio's asset allocation. I am a 30-year-old male with \$250,000 in my retirement fund. I like to invest on the aggressive side because I hope to retire early. Please let me know what percentage I should invest in stocks, bonds and cash? Provide the advice in a professional tone. Place the suggested allocation in a table format and provide a column with the reasoning behind that percentage allocation.

Acting as a financial advisor, provide a breakdown for my financial portfolio's asset allocation. I am a 30 year old male with \$250,000 in my retirement fund. I like to invest on the aggressive side because I hope to retire early. Please let me know what percentage I should invest in stocks, bonds and cash? Provide the advice in a professional tone. Place the suggested allocation in a table format and provide a column with the reasoning behind that percentage allocation.

Certainly! Given your age, investment horizon, and preference for an aggressive investment strategy, here's a suggested asset allocation for your retirement portfolio:

Asset Class	Percentage	Reasoning
Stocks	85%	Stocks offer the highest potential for growth, which aligns with your long-term goal of early retirement. Given your age, you have a longer time horizon to ride out market volatility.
Bonds	10%	Bonds provide some stability and income, helping to balance the higher risk associated with a predominantly stock portfolio. This allocation helps mitigate potential losses during market downturns.
Cash	5%	Maintaining a small cash reserve ensures liquidity for emergencies or short-term needs without having to sell investments at an inopportune time.

This allocation aims to maximize growth while maintaining a level of diversification to manage risk. Regularly reviewing and rebalancing your portfolio is essential to ensure it continues to align with your financial goals and risk tolerance ¹ ² ³.

By using the strategies and giving more detail, our result is much more suitable for our purpose.